

Kanban_BPS#56 — Peer Skill Exchange & Mentorship Network

Header Identifier: Nishanth T Chigilar_PES1UG24CS302-BPS#56

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- Problem Statement: #56 — Peer Skill Exchange & Mentorship Network
- Course: Software Engineering Lab (UE24CS252AA) — Lab 2: Agile Project Management (Kanban Simulation)
- Project Key: KBPS56

1. Objective & Kanban Methodology

The objective of this project is to implement an Agile Kanban framework using Atlassian Jira to manage continuous delivery for the Peer Skill Exchange & Mentorship Network. While Scrum operates on fixed-length, time-boxed sprint iterations, Kanban focuses on continuous workflow visualization, reducing bottlenecks, and managing Work-In-Progress (WIP).

Requirements identified in Lab 1 are structured as actionable user stories linked to parent functional Epics, broken down into executable child subtasks, and progressed across workflow states based on team capacity.

2. Requirement Engineering & Epic Mapping

To organize the technical scope of the peer skill exchange platform, requirements are grouped under three parent Epics, spanning five Functional Requirements (FRs) and two Non-Functional Requirements (NFRs):

- **Epic 1: Time-Banking Ledger & Credit Transactions:** Manages non-monetary credit accounting, temporary escrow locking, and automated transfers upon dual verification.
- **Epic 2: Mentor Discovery & Session Scheduling:** Encompasses mentor skill tagging, multi-parameter search filtering, slot reservation, and external calendar synchronization.
- **Epic 3: Session Validation, Feedback & Disputes:** Manages mutual session completion sign-offs, 1–5 star reviews with qualitative feedback, and dispute escalation handling.

Requirements Inventory (Stories)

Issue Key	Requirement Type	Summary / User Story	Parent Epic	Workflow State
KBPS56-4	Functional (FR-001)	Automated Credit Transfer: As a Skill Mentor, I want 1 time credit transferred to my balance upon session verification, so that my teaching time is fairly rewarded.	Epic 1: Ti... ▾	In Progress ▾
KBPS56-5	Functional (FR-002)	Search & Filter Mentors: As a Learner Member, I want to search and filter mentor profiles by skill tags, ratings, and open slots, so that I can easily find a suitable mentor.	Epic 2: M... ▾	Done ▾
KBPS56-6	Functional (FR-003)	Session Booking & Escrow Lock: As a Learner Member, I want to book an available mentor slot by holding 1 credit	Epic 2: M... ▾	In Progress ▾

Issue Key	Requirement Type	Summary / User Story	Parent Epic	Workflow State
		in escrow, so that my session is guaranteed without immediate deduction.		
KBPS56-7	Functional (FR-004)	Dual Completion Validation & Rating: As a Learner Member, I want to confirm session completion and submit a 1–5 star rating with comments within 48 hours, so that the mentor receives credit and the community sees verified feedback.	Epic 3: S... ▾	Done ▾
KBPS56-8	Functional (FR-005)	Cancellation & Dispute Handling: As a Community Member, I want to cancel sessions >24 hours prior for a full refund or raise a dispute	Epic 3: S... ▾	To Do ▾

Issue Key	Requirement Type	Summary / User Story	Parent Epic	Workflow State
		for a no-show, so that my time credits are protected.		
KBPS56-9	Non-Functional (NFR-001)	External Calendar Sync: As a Learner Member, I want confirmed sessions to export an .ics feed to Google and Outlook calendars within 5 seconds, so that my external schedule stays up to date.	Epic 2: M... ▾	To Do ▾
KBPS56-10	Non-Functional (NFR-002)	ACID Ledger Integrity & Encryption: As a Platform Admin, I want all credit ledger transactions to execute with strict ACID consistency and TLS/AES encryption, so that credit balances remain tamper-proof.	Epic 1: Ti... ▾	To Do ▾

3. Kanban Board Setup & Workflow Distribution

The Jira Kanban board is partitioned into three core status columns representing the operational pipeline: **To Do**, **In Progress**, and **Done**. Work items transition across these stages as engineering capacity permits.

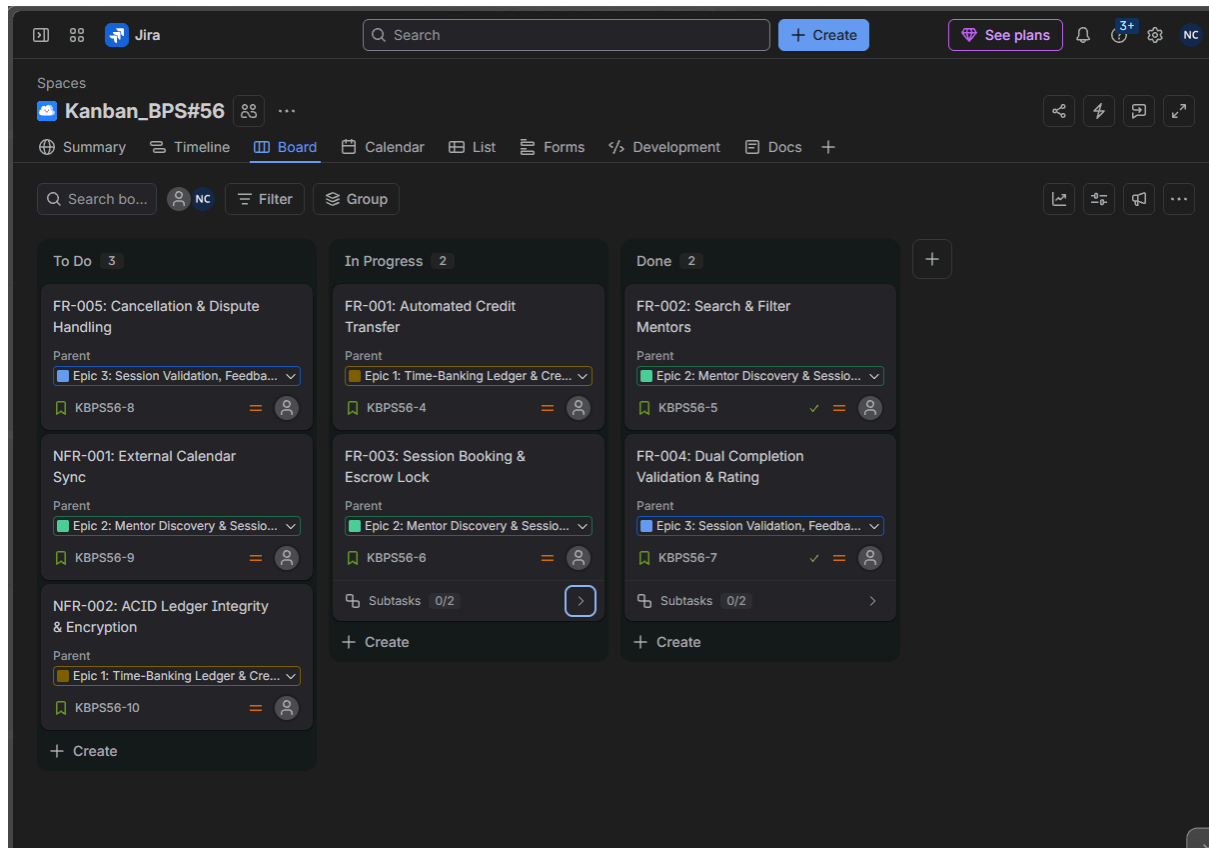


Figure 1: Jira Kanban Board (Kanban_BPS#56) displaying all 7 requirement stories distributed across To Do (3), In Progress (2), and Done (2) with color-coded parent Epic badges and subtask indicators.

Column Workflow Analysis

- **To Do (3 Work Items):** Contains requirements awaiting development bandwidth (KBPS56-8: Cancellation & Dispute Handling, KBPS56-9: External Calendar Sync, and KBPS56-10: ACID Ledger Integrity & Encryption). These items reside in the queue until active development slots open.
- **In Progress (2 Work Items):** Visualizes current active development (KBPS56-4: Automated Credit Transfer and KBPS56-6: Session Booking & Escrow Lock). Limiting concurrent development to 2 core transaction items prevents context-switching and enforces Work-In-Progress discipline.
- **Done (2 Work Items):** Reflects completed, verified, and accepted deliverables ready for deployment (KBPS56-5: Search & Filter Mentors and KBPS56-7: Dual Completion Validation & Rating).

4. Granular Task Decomposition & Child Subtasks

To prevent bottlenecks and provide visibility into technical execution, complex requirement cards are broken down into child subtasks. This structure allows developers to verify incremental prerequisites before advancing the parent story across the board.

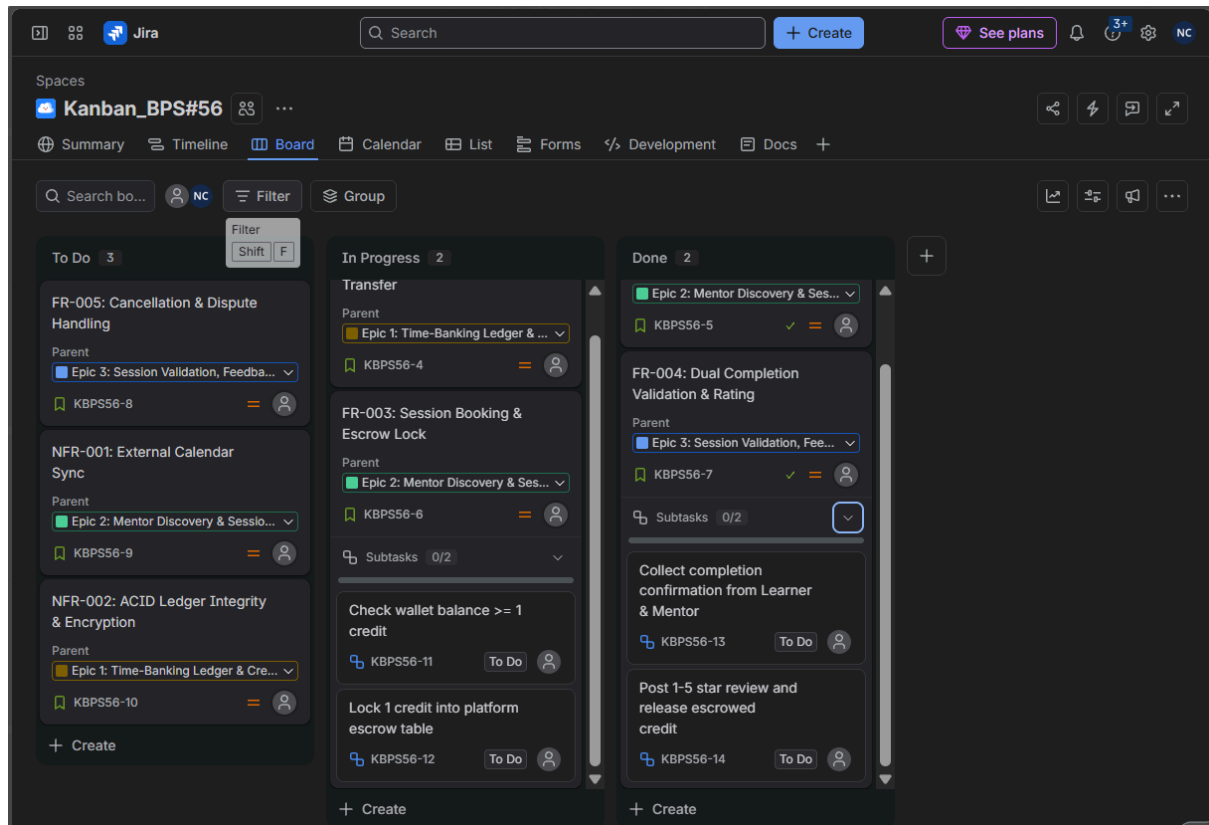


Figure 2: Jira issue detail drawers showcasing granular child subtasks configured under FR-003: Session Booking & Escrow Lock and FR-004: Dual Completion Validation & Rating.

Detailed Subtask Specifications

1. FR-003: Session Booking & Escrow Lock (KBPS56-6)

- **Parent Epic:** Epic 2: Mentor Discovery & Session Scheduling
- **Status:** In Progress
- **Subtask 1 (KBPS56-11):** Check wallet balance ≥ 1 credit — Pre-checkout conditional check querying the user ledger to ensure sufficient non-monetary balance prior to slot reservation.
- **Subtask 2 (KBPS56-12):** Lock 1 credit into platform escrow table — Database row transaction that temporarily debits available balance and binds 1 credit into an escrow holding state until dual verification occurs.

2. FR-004: Dual Completion Validation & Rating (KBPS56-7)

- **Parent Epic:** Epic 3: Session Validation, Feedback & Disputes

- **Status:** Done
- **Subtask 1 (KBPS56-13):** Collect completion confirmation from Learner & Mentor — Two-way handshake verification ensuring both participants acknowledge session delivery within the 48-hour window.
- **Subtask 2 (KBPS56-14):** Post 1-5 star review and release escrowed credit — Commits reviewer scores and qualitative comments to the mentor profile while triggering an automated event to transfer 1 credit from escrow to the mentor's available balance.

5. Summary & Engineering Outcomes

- **Traceability:** Every technical requirement from Lab 1 directly maps to a Jira Story linked to a functional Epic, establishing clear requirement traceability.
- **Decomposition:** High-level user stories deconstruct into atomic database and UI subtasks, enabling clear progression tracking without task ambiguity.
- **Flow Efficiency:** Visualizing work states on the Kanban board highlights active operational dependencies, preventing over-commitment and ensuring consistent task delivery.